A Technical Guide to the Physicochemical Properties of Phenkapton (CAS 2275-14-1)

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Compound of Interest

Compound Name:	*Phenkapton*
CAS No.:	*2275-14-1*
Cat. No.:	*B1204249*

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Abstract: This technical guide provides a comprehensive overview of the physicochemical properties of **Phenkapton** (CAS No. 2275-14-1), an organothiophosphate pesticide. Designed for researchers, chemists, and environmental scientists, this document synthesizes available data on the compound's chemical identity, physical characteristics, stability, and analytical methodologies. By explaining the causality behind its properties and detailing protocols for its characterization, this guide serves as an essential resource for professionals working with this compound. All data and claims are supported by authoritative references to ensure scientific integrity.

Chemical Identity and Molecular Structure

Phenkapton is an organophosphorus compound primarily classified as a pesticide and acaricide.[1][2] Its chemical identity is defined by a unique combination of a dichlorophenyl ring linked via a thioether methyl bridge to a phosphorodithioate core.[1] This structure dictates its physicochemical behavior, including its high lipophilicity and mechanism of action as an acetylcholinesterase inhibitor.[1][3]

Table 1: Chemical Identifiers for **Phenkapton**

Identifier	Value	Source(s)
CAS Number	**2275-14-1**	**[1][4][5]**
IUPAC Name	S-{[(2,5-dichlorophenyl)sulfanyl]methyl} O,O-diethyl phosphorodithioate	[1][2]
Molecular Formula	$C_{11}H_{15}Cl_2O_2PS_3$	[1][4][5]
Molecular Weight	377.31 g/mol	[4][5][6]
Canonical SMILES	CCOP(=S)(OCC)SCSC1=C(C=CC(=C1)Cl)Cl	[1][5]
InChI Key	GGNLTHFTYNDYNK-UHFFFAOYSA-N	[1][4]

| Synonyms | Phencapton, G 28029, ENT 25585, Phenudin |[4][5][6] |

The molecule's structure features a rigid dichlorophenyl aromatic ring, which contributes to its environmental persistence, and a flexible thiomethyl linker connecting to the reactive phosphorodithioate group.[1] The molecule possesses five hydrogen bond acceptor sites but no donors, a characteristic that influences its solubility and intermolecular interactions.[1][5]

Caption: 2D structure of **Phenkapton** (CAS 2275-14-1).

Physical Properties

The physical state and solubility characteristics of **Phenkapton** are critical for its application, environmental distribution, and analytical extraction. The data, summarized in Table 2, reflects its non-polar, lipophilic nature.

Table 2: Summary of Physicochemical Data for **Phenkapton**

Property	Value	Comments and Source(s)
Physical Form	**White powder**	**Also available as a clear, colorless liquid when in solution (e.g., acetonitrile).[7][8]**
Melting Point	~50-52 °C	One source reports this range, while others state data is not available.[1][8]
Boiling Point	120 °C at 1 x 10^{-3} Torr ~250 °C (est.) 409.7 ± 55.0 °C (est. at 760 mmHg)	Boiling point is highly dependent on pressure. The value at reduced pressure is from experimental data.[1][4]
Water Solubility	0.044 mg/L at 20 °C	Very low solubility, characteristic of lipophilic organophosphates.[1][3]
Organic Solubility	Readily soluble	High solubility in solvents like acetonitrile, chloroform, and cyclohexane.[1]
Density	1.440 g/cm³	[8]

| LogP (Octanol-Water) | 5.85 | Indicates extremely high lipophilicity and bioaccumulation potential.[1][3][5] |

Solubility Profile: Causality and Implications

Phenkapton's very low water solubility is a direct consequence of its molecular structure.[1] The large, non-polar 2,5-dichlorophenyl group and the diethyl ester moieties dominate the molecule's character, creating a hydrophobic entity that resists interaction with polar water molecules.[1] Conversely, these features promote high solubility in non-polar and moderately polar organic solvents.[1]

This solubility profile has significant real-world implications:

- Environmental Fate: In aquatic systems, **Phenkapton** is more likely to adsorb to sediment and suspended organic matter rather than remain dissolved in the water column.[1]
- Bioavailability & Bioaccumulation: Its high octanol-water partition coefficient (LogP) of ~5.85 suggests a strong tendency to partition into fatty tissues of organisms, leading to a high potential for bioaccumulation.[1]
- Analytical Chemistry: The high solubility in organic solvents facilitates its extraction from complex matrices (e.g., soil, tissue) during residue analysis.[1]

Chemical Stability and Reactivity

Understanding the stability of **Phenkapton** is crucial for storage, handling, and predicting its environmental persistence.

- Thermal Stability: **Phenkapton** shows decreased stability at elevated temperatures, with significant decomposition reported above 200°C.[1] Thermal degradation can release toxic gases, including phosphorus oxides and chlorides.[1][9]
- Chemical Reactivity: As with other organophosphates, **Phenkapton** is susceptible to reaction with strong reducing agents, which can produce highly toxic phosphine gas.[1][9] While not rapidly reactive with air or water under standard conditions, partial oxidation can yield toxic byproducts.[1][9]
- Environmental Persistence: In aquatic environments, **Phenkapton** is considered highly persistent.[1] Its low water solubility and resistance to hydrolysis contribute to its long-lasting presence and potential for bioaccumulation in aquatic food chains.[1]

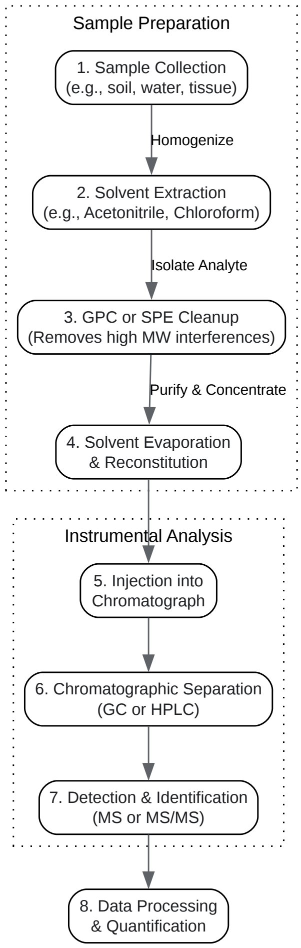
Spectroscopic and Analytical Characterization

Accurate identification and quantification of **Phenkapton** require modern analytical techniques. The primary methods involve chromatography coupled with mass spectrometry.

Chromatographic Analysis

Gas chromatography (GC) and high-performance liquid chromatography (HPLC) are the principal methods for separating **Phenkapton** from complex sample matrices.[10][11]

- Gas Chromatography (GC): Often coupled with a flame photometric detector (FPD) in phosphorus mode for selectivity, or with a mass spectrometer (MS) for definitive identification.[5][10]
- High-Performance Liquid Chromatography (HPLC): Typically coupled with tandem mass spectrometry (MS/MS), this technique offers high sensitivity and specificity for residue analysis in various products.[10][11]



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Caption: General workflow for the analysis of **Phenkapton** residues.

Protocol 1: General Procedure for Pesticide Residue Analysis via GC/MS

This protocol describes a generalized, self-validating system for determining **Phenkapton** levels in a solid matrix like soil. The causality for each step is explained to provide field-proven insight.

- Sample Preparation & Extraction:
  - a. Weigh 10 g of a homogenized soil sample into a centrifuge tube.
  - b. Add 20 mL of acetonitrile. Rationale: Acetonitrile is a polar aprotic solvent effective at extracting a wide range of pesticides, including organophosphates, from the soil matrix.
  - c. Fortify with an appropriate internal standard (e.g., D10-**Phenkapton**) to correct for extraction efficiency and instrument variability.[12]
  - d. Vortex for 2 minutes, then sonicate for 15 minutes to ensure thorough extraction.
  - e. Centrifuge at 4000 rpm for 10 minutes to separate the solid and liquid phases.
  - f. Carefully transfer the supernatant (acetonitrile extract) to a clean flask.
- Extract Cleanup (Size Exclusion):
  - a. Concentrate the extract to 1-2 mL using a rotary evaporator or nitrogen stream. Rationale: Concentration is necessary to meet the detection limits of the instrument.
  - b. Perform Gel Permeation Chromatography (GPC) cleanup to remove high-molecular-weight interferences like humic acids.[13] Rationale: GPC separates molecules by size, effectively removing large, interfering compounds that can contaminate the GC system and cause ion suppression in the MS source.
- GC/MS Analysis:
  - a. Inject 1 µL of the cleaned extract into a GC/MS system.
  - b. GC Conditions (Typical): Use a non-polar capillary column (e.g., DB-5ms). Start with an oven temperature of 80°C, hold for 1 min, then ramp to 280°C at 10°C/min, and hold for 5 min. Rationale: This temperature program allows for the separation of a wide range of semi-volatile compounds.

- c. MS Conditions: Operate in Electron Ionization (EI) mode. Use Selected Ion Monitoring (SIM) for enhanced sensitivity, monitoring for characteristic **Phenkapton** ions. Rationale: SIM mode increases the signal-to-noise ratio by focusing only on ions specific to the target analyte, allowing for lower detection limits.

- Validation and Quantification:
  - a. Run a multi-point calibration curve using certified **Phenkapton** standards to establish linearity and response factors.
  - b. Analyze a procedural blank and a fortified matrix spike alongside samples to check for contamination and recovery.
  - c. Confirm the identity of **Phenkapton** by comparing the retention time and the ratio of quantifier/qualifier ions to that of a certified standard.

Predicted Spectroscopic Signatures

While publicly available NMR and IR spectra for **Phenkapton** are scarce, its structure allows for the prediction of key spectroscopic features. This is a fundamental exercise in structure elucidation.

Nuclear Magnetic Resonance (NMR) Spectroscopy:

- ^{1}H NMR: Expected signals would include triplets and quartets in the 1.0-1.5 ppm and 3.9-4.3 ppm regions, respectively, corresponding to the two ethoxy groups ($-OCH_2CH_3$). A singlet for the thiomethyl bridge ($-SCH_2S-$) would likely appear around 4.5-5.0 ppm. The aromatic protons on the dichlorophenyl ring would produce complex multiplets in the 7.0-7.5 ppm region.

- ^{13}C NMR: Distinct signals would be expected for the methyl (~15 ppm) and methylene (~65 ppm) carbons of the ethoxy groups, the thiomethyl bridge carbon (~40 ppm), and multiple signals in the aromatic region (120-140 ppm).

Infrared (IR) Spectroscopy: Key absorption bands would be predicted for:

- P=S stretch: A strong band around 650-850 cm^{-1}.

- P-O-C stretch: Strong absorptions around 950-1050 cm^{-1}.
- Aromatic C=C stretch: Medium intensity peaks in the 1450-1600 cm^{-1} region.
- C-H stretches: Aliphatic C-H stretches just below 3000 cm^{-1} and aromatic C-H stretches just above 3000 cm^{-1}.
- C-Cl stretch: Strong bands in the 600-800 cm^{-1} region.

Conclusion

Phenkapton is a highly lipophilic organothiophosphate with very low water solubility and a high potential for bioaccumulation. Its physicochemical properties are dictated by its dichlorophenyl and phosphorodithioate functional groups. The compound is thermally sensitive and requires careful handling. Standard analytical workflows for its detection and quantification involve solvent extraction followed by chromatographic separation and mass spectrometric detection. This guide provides the foundational data and procedural insights necessary for scientists and researchers engaged in the study, use, or environmental monitoring of this compound.

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